Taylor expansion commutes with approximation trees in distant Bang calculus

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Editorial note. Source https://arxiv.org/abs/2601.05199v4. Copyright Kostia Chardonnet, Jules Chouquet and Axel Kerinec, CC BY4.0. The main Taylor/approximation-tree commutation theorem and full author proof continuation are preserved. FSCD2026.11 belongs to the same work. Adjacent translation/meaningfulness material is retained for context and original labels, with no extra counted claims.

Allowed standard prerequisites (editorial).

- Lambda binding/substitution, finite multisets, reduction normal forms and directed approximation ideals. Native distant-Bang section. Actual calculus-specific resource normalisation, Taylor simulation and approximant reconstruction proofs are retained.

2 The (distance) Bang-calculus

We begin by recalling the theory of `dBang`, first with some results from previous studies [18, 5, 11], before developing its approximation theory via Taylor expansion and approximation trees.

Definition 1. (`dBang`: *terms and contexts)*

(Terms)	$M, N := x \mid MN \mid \lambda x M \mid !M \mid \mathbf{der}(M) \mid M[N/x]$
(List contexts)	$L := \Box \mid L[M/x]$
(Surface contexts)	$S := \Box \mid SM \mid MS \mid \lambda x S \mid \mathbf{der}(S) \mid S[M/x] \mid M[S/x]$
(Full contexts)	$F := \Box \mid FM \mid MF \mid \lambda x F \mid \mathbf{der}(F) \mid F[M/x] \mid M[F/x] \mid !F$

Terms include the standard λ-calculus constructs: variable (ranging over a countably infinite set), application and abstraction. Two additional ones are used to define the `Bang`-calculus: the *bang* (or *exponential*) $!M$, representing delayed evaluation of the subterm M, and *dereliction* $\mathbf{der}(M)$, which reactivates it. Finally, *explicit substitutions* $M[N/x]$ comes from the "at distance" mechanism, and represents pending substitutions. The lambda-abstraction (as usual) and the explicit substitution bind the variable x in M. We use *contexts*, *i.e.* terms with a subterm hole ($\Box$) and we denote by $C\langle M\rangle$ the term obtained by filling the hole in C by M. We distinguish *list, surface* and *full* contexts. List contexts are sequences of explicit substitutions and will be used for the reduction at a distance. Surface and full contexts determine whether reduction under a ! is allowed (in particular it is forbidden in *weak* calculi (`CbN` or `CbV`)).

The `dBang` calculus has three reduction rules:

$$L\langle \lambda x M\rangle N \mapsto_! L\langle M[N/x]\rangle M[L\langle !N\rangle/x] \mapsto_! L\langle M\{N/x\}\rangle \mathbf{der}(L\langle !M\rangle) \mapsto_! L\langle M\rangle$$

Notice that requiring subterms of the form $!M$ assigns them *value* status in the `CbV` sense.

We write $\to_{!s}$ and $\to_!$ for the closures under surface and full contexts, respectively; both are confluent [18, Theorem 1]. We also denote by $\to_i$ the internal reductions (*i.e.* $\to_i = \to_! \setminus \to_{!s}$, these are the reductions occurring under a !-construct).

Example 2.

$$\begin{aligned}
&(\lambda xxy)[!z/y]!!((\lambda ww)!N)\\
&\to_{!s} (xy)[!!((\lambda ww)!N)/x][!z/y]\\
&\to_{!s} !((\lambda ww)!N)y[!z/y]\\
&\to_{!s} !((\lambda ww)!N)z\\
&\to_{!s} !(w[!N/w])z\\
&\to_i \quad !Nz
\end{aligned}$$

Example 3.

- $\Delta = \lambda x(x!x), \Omega = \Delta!\Delta$. *We have* $\Omega \to_{!s}^{2} \Omega$.
- $Y_x^n = (\lambda yx!(y!y))!(\lambda yx!(y!y))$. *We have* $Y_x^n \to_{!s}^{+} x!Y_x^n$
- $Y_x^v = (\lambda yx(y!y))!(\lambda yx(y!y))$. *We have* $Y_x^v \to_{!s}^{+} xY_x^v$

The upper scripts in the two last items, as we shall see further, represent the fact that Y_x^n *and* Y_x^v *correspond respectively to the* `CbN` *and* `CbV` *translations from a fixpoint combinator* Y *of standard* λ*-calculus*[1].

2.1 Meaningfulness in `dBang`

This section focus on *meaningfulness*, later related to approximation theory in Section 4: intuitively, a term is meaningful if it reduces to a desired result under some *testing context*. It generalizes the notions of *solvability* `CbN` and *scrutability* `CbV`, and has been studied in detail for `dBang` in [18].

Definition 4. *A term* M *of* `dBang` *is said* meaningful *if there exists a testing context* $T := \Box \mid TM \mid (\lambda xT)M$ *and a term* P *such that* $T\langle M\rangle \to_{!s}^{*} !P$.

Notice that the surface reduction involved in meaningfulness is not restrictive: for any N, $N \to_{!}^{*} !P$, then there is some P' such that $N \to_{!s}^{*} !P'$ (this follows from a standardisation property, see Corollary 6 below). It has been shown in [18] that the smallest theory that identifies all meaningless terms is consistent; that meaningful terms and surface-normalizing terms can be characterized by an intersection type systems and finally that meaningless subterms do not affect the operational meaning of a given term [18, Proposition 8, Theorem 24 and Corollary 11].

A natural property of surface reduction is that it determines the external shape of a term. In other words, allowing full reduction does not unlock external redexes. Indeed, with terms like *e.g.* $\lambda x!M$, $y!M$, $!M!M$, reductions occurring contextually inside M cannot make the full term reducible at the surface (the syntax of such term will be precisely characterized further). This is expressed as the following factorization proposition:

Proposition 5. *([5], Corollary 21) Let* $M \to_{!}^{*} N$. *There is some* P *such that* $M \to_{!s}^{*} P \to_{i}^{*} N$.

Conversely, internal reductions preserve the *external shape*. We express this notion with multi-holes surface contexts, that let us reformulate the factorization property as follows:

Corollary 6 (Standardization)**.** *Let* S^+ *denote multi-holes surface contexts and given by the syntax:* $S^+ := \Box \mid M \mid S^+S^+ \mid S^+[S^+/x] \mid \mathbf{der}(S^+) \mid \lambda xS^+$. *For any reduction* $M \to_{!}^{*} P$, *there are some terms* N_i *and a multi-hole surface context* S *such that* $P = S\langle !N_1, \dots, !N_k\rangle$ *and* $M \to_{!s}^{*} S\langle !N_1', \dots, !N_k'\rangle \to_{i}^{*} S\langle !N_1, \dots, !N_k\rangle$

If $k = 0$, the context has no hole and the reduction occurs only at surface level.

2.2 Taylor expansion

We define the Taylor expansion of `dBang`, starting with the associated resource calculus δ`Bang`. This calculus will also serve as the resource language for `dCbN` and `dCbV` in Section 3. There is no necessity to define specific resource calculi, as δ`Bang` fits well as a target of usual Taylor expansion (`CbN` [16] and cbv [14]), with a straightforward adaptation to distant setting.

2.2.1 `dBang`: resources

Definition 7 (Resource calculus δ`Bang`)**.**

(terms)	$m, n := x \mid mn \mid \lambda xm \mid \mathbf{der}(m) \mid m[n/x] \mid [m_1, \dots, m_k]$
(lists)	$l := \Box \mid l[m/x]$
(surface)	$s := \Box \mid sm \mid ms \mid \lambda xs \mid \mathbf{der}(s) \mid s[m/x] \mid m[s/x]$
(full)	$f := \Box \mid fm \mid mf \mid \lambda xf \mid \mathbf{der}(f) \mid f[m/x] \mid m[f/x] \mid [f, m_1, \dots, m_k]$
(tests)	$t := \Box \mid tm \mid (\lambda xt)m$

[1] Note that Y_x^v lacks some good properties if one would want to use it for computations in `CbV`, and `CbV` fixpoints are usually defined differently. We nevertheless keep Y_x^v as a running example, so as to illustrate an infinite computation, with a fixpoint behaviour, but with empty semantics.

$$\frac{}{x \lhd_! x} \qquad \frac{m \lhd_! M}{\lambda x m \lhd_! \lambda x M} \qquad \frac{m \lhd_! M}{\mathbf{der}(m) \lhd_! \mathbf{der}(M)} \qquad \frac{m \lhd_! M \qquad n \lhd_! N}{mn \lhd_! MN}$$

$$\frac{m \lhd_! M \qquad n \lhd_! N}{m[n/x] \lhd_! M[N/x]} \qquad \frac{m_1 \lhd_! M \qquad \dots \qquad m_k \lhd_! M}{[m_1, \dots, m_k] \lhd_! !M}\ k \in \mathbb{N}$$

Figure 1: Resource approximation for dBang

Notations (resource terms, multisets/bags, multilinear substitutions) Terms $[m_1, \dots, m_k]$ for $k \in \mathbb{N}$, called *bags*, denote finite multisets of resource terms, with $[\,]$ the empty bag. When necessary, we use a subscript as $[m]_k$ or $[m, \dots, m]_k$ in order to make explicit the number of occurrences.

Contexts are exactly as in dBang (Definition 1), with bags replacing exponentials. We write P_k for the sets of permutations of the set $\{1, \dots, k\}$, and we denote as $d_x(m)$ the number of free occurrences of the variable x in m. Resource substitution is (multi-)linear, and ordered: when we write $m\{n_1/x_1, \dots, n_k/x_k\}$, it is always intended (in the resource setting) that x_i represents the i-th free occurrence of x in m. In that way, each term n_i is substituted exactly once in m.

We are now ready to define the reduction relation.

Definition 8.

The reduction relation $\Rightarrow_\delta \subseteq \delta\texttt{Bang} \times \wp(\delta\texttt{Bang})$ is then defined as follows:

- $l\langle \lambda x m\rangle n \Rightarrow_\delta \{l\langle m[n/x]\rangle\}$
- $m[l\langle [n_1, \dots, n_k]\rangle / x] \Rightarrow_\delta \begin{cases} \bigcup_{\sigma \in P_k} l\langle m\{n_{\sigma(1)}/x_1, \dots, n_{\sigma(k)}/x_k\}\rangle & \text{if } k = d_x(m) \\ \emptyset & \text{otherwise} \end{cases}$
- $\mathbf{der}(l\langle [m_1, \dots, m_k]\rangle) \Rightarrow_\delta \{l\langle m_1\rangle\}$ *if* $k = 1$ *and* $\emptyset$ *otherwise*

We write $m \to_\delta n$ as soon as $m \Rightarrow_\delta X$ and $n \in X$ for some n (if $m \Rightarrow_\delta \emptyset$, we also abusively write $m \to_\delta \emptyset$, and we add the following equation: if $f\langle \emptyset\rangle = \emptyset$ for any full context f). We also define $\to_{\delta s}$ and $\to_\delta$ the contextual closures of $\to_\delta$ under surface and full contexts, respectively. Notice that none of these reductions is deterministic. Both $\to_{\delta s}$ and $\to_\delta$ are strongly normalizing, which is an immediate consequence of linearity: the size of bags of resource terms is decreasing. Confluence of $\to_{\delta s}$ and $\to_\delta$ follows from standard resource-calculus arguments and from the proofs for $\to_{!s}$ and $\to_!$.

2.2.2 Approximation

In Figure 1, defines the relation $\lhd_! \subseteq \delta\texttt{Bang} \times \texttt{dBang}$, where $m \lhd_! M$ means that m is a multilinear resource approximation on M.

We extend this definition to list contexts as follows: $\Box \lhd_! \Box$, $l[m/x] \lhd_! L[M/x]$ if $l \lhd_! L$ and $m \lhd_! M$. The extension to surface and full contexts follows analogously.

The expected behaviour of context approximation is given by the following lemma (proved by induction on contexts):

Lemma 9.

- *If $m \lhd_! L\langle N\rangle$, then there exist $l \lhd_! L$ and $n \lhd_! N$ such that $m = l\langle n\rangle$.*
- *If $m \lhd_! S\langle N\rangle$, then there exist $s \lhd_! S$ and $n \lhd_! N$ such that $m = s\langle n\rangle$.*

Notice that this property fails for full contexts. Indeed, consider $t = [] \lhd_! !M = F\langle M\rangle$, there is no $m \lhd_! M$ or $f \lhd_! F$ such that $m \lhd_! M$ and $t = f\langle m\rangle$. Consequently, our definitions do not provide a convenient notion of approximation between full contexts. This is due to a need for parallel treatment of bags, which - as we shall see later - is incompatible with single-hole contexts.

Definition 10. *(Taylor expansion) For any $M \in \texttt{dBang}$, we define its Taylor expansion as the set of its resource approximants:*

$$\mathcal{T}(M) = \{m \in \delta\texttt{Bang} \mid m \lhd_! M\}$$

By strong normalization of δBang, we can define the normal form $\mathbf{nf}(m)$ of a resource term m as the finite set made of its full reducts. Remember that the resource reduction we consider is not deterministic ($\Rightarrow_\delta$ is, but not $\to_\delta$), hence this set can contain multiple terms, and can be empty. We then define the *Taylor normal form* of dBang terms as $\mathrm{TNF}(M) = \bigcup_{m \lhd_! M} \mathbf{nf}(m)$. Notice that Taylor normal form is made of full normal terms, not only surface normal forms.

Remark 11. *If $\mathcal{T}(M) \Rightarrow_\delta^* X$ and X is a set of full normal terms, than $X = TNF(M)$. But the converse is not true: we can have $TNF(M)$ non empty, but no k such that $M \Rightarrow_\delta^k TNF(M)$. This is due to the fact that terms in $\mathcal{T}(M)$ can require different number of reduction steps, as with Taylor expansion of fixpoint combinators where this number is unbounded, as illustrated in Example 12.*

Example 12. *Consider the terms given in Example 3.*

- *An approximant of $m \lhd_! \Omega$ must be of shape $(\lambda xx[x]_k)[\lambda xx[x]_{k_1}, \dots, \lambda xx[x]_{k_l}]$. Now, if $k = l - 1$ (otherwise $m \mapsto_! \emptyset$) then $m \to_{\delta s} (\lambda xx[x]_{k_1})[\lambda xx[x]_{k_2}, \dots, \lambda xx[x]_{k_l}]$[2], which is again an approximant of Ω. But we can observe that the cardinality of the bag reduces during this reduction; hence if we iterate this reduction, we eventually reach a term like $(\lambda x[x]_k)[] \to_{\delta s} \emptyset$ (if an empty reduction has not occurred before). So, $TNF(\Omega) = \emptyset$.*
- *Similarly, if $m \lhd_! Y_x^n$, we verify easily that $m \to_{\delta s} x[n_1, \dots, n_k]$, with $n_i \lhd_! Y_x^n$. In particular, $x[] \lhd_! Y_x^n$ and is in normal form. Actually, $TNF(Y_x^n)$ can be characterized inductively : $x[] \in TNF(Y_x^n)$, and if $n_1, \dots, n_k \in TNF(Y_x^n)$, then $x[n_1, \dots, n_k] \in TNF(Y_x^n)$, for any k.*
- *The other fixpoint term, Y_x^v, behaves slightly differently: if $m \lhd_! Y_x^v$, then we verify $m \to_{\delta s}^* xn$, for some $n \lhd_! Y_x^v$. But here, because of the argument not being in a bag, all approximants reduce (if not $\emptyset$) to some term $xxx \dots xn$, but are not in normal form; since such a reduction terminates, we observe that $TNF(Y_x^v) = \emptyset$.*

Remark 13. *[Clashes and normal forms] Note that we have not excluded so-called* clashes *from our calculus, as they cause no issue in our setting. Thus, terms such $\mathbf{der}(\lambda xx)$ are considered as regular normal forms. This approach is similar to the one of Dufour and Mazza [13]. However, giving an empty semantics to clashes could easily be done, for example, adding reductions such as $\mathbf{der}(\lambda xm) \to \emptyset$ to resource calculus.*

2.2.3 Simulation

We now formalize the simulation of `dBang` reduction in the approximants of δ`Bang`, as sketched in the Example 12.

Lemma 14 (Substitution Lemma for Taylor expansion)**.** *For any M, N of* `dBang`*, for any $m, n_1, \dots, n_{d_x(m)}$ of δ`Bang`, we have $m\{n_1/x_1 \dots n_{d_x(m)}/x_{d_x(m)}\} \lhd_! M\{N/x\}$ if and only if $m \lhd_! M$, and $n_i \lhd_! N_i$ for all $i \leq d_x(m)$.*

We can show, by a straightforward induction, with the help of the Substitution Lemma stated above, that the surface reduction acts exactly the same way in a `dBang` term and in its approximants:

Lemma 15. *Let $m \lhd_! M$ and $m \to_{\delta s} n$. Then there is $N \in$ `dBang` such that $M \to_{!s} N$ and $n \lhd_! N$.*

However, this result is false when considering reduction in full contexts: take the following term $m = [(\lambda xx)y, (\lambda xx)y]$. We have $m \lhd_! !((\lambda xx)y)$ and $m \to_\delta n = [x[y/x], (\lambda xx)y]$, but n is not an approximant of any `dBang` term (it has lost its uniformity, in the sense of Ehrhard and Regnier [16]). We will need parallel reduction to achieve full simulation, this is done in Subsection 2.2.4.

Lemma 16. *Considering the following configuration:*

$$\begin{array}{ccc} M & \to_{!s} & N \\ \triangledown & & \triangledown \\ m & \to_{\delta s} & n \end{array}$$

Given three of four terms, we can always obtain a convenient fourth that makes the square commute. (for a proof, see lemmas 74, 75 and 76 in Appendix A).

Definition 17. *Let X and Y be sets, and $\mathcal{R} \subseteq X \times Y$ a relation. We write $X \overline{\mathcal{R}} Y$ whenever $\forall x \in X$, $\exists y \in Y$ such that $(x, y) \in \mathcal{R}$ and $\forall y \in Y$, $\exists x \in X$ such that $(x, y) \in \mathcal{R}$.*

Then, using Lemma 16 we obtain a simulation theorem for surface reduction:

Theorem 18 (Simulation)**.** *Let M, N in* `dBang` *such that $M \to_{!s} N$. We have $\mathcal{T}(M) \overline{\Rightarrow_{\delta s}} \mathcal{T}(N)$.*

[2]We consider here one possible reduction, any element of the bag could be substituted to the inner head variable x, not necessarily $\lambda x[x]_{k_1}$, the argument is valid for all the reduction paths.

$$\frac{}{x \rightrightarrows_\delta x} \qquad \frac{m_1 \rightrightarrows_\delta m'_1 \quad \dots \quad m_k \rightrightarrows_\delta m'_k}{[m_1,\dots,m_k] \rightrightarrows_\delta [m'_1,\dots,m'_k]}\ k \in \mathbb{N}$$

$$\frac{m \rightrightarrows_\delta m'}{s\langle m\rangle \rightrightarrows_\delta s\langle m'\rangle}\ * \qquad \frac{[m_1,\dots,m_k] \rightrightarrows_\delta [m'_1,\dots,m'_k] \qquad n \rightrightarrows_\delta n' \qquad l \rightrightarrows_\delta l'}{n[l\langle[m_1,\dots,m_k]\rangle] \rightrightarrows_\delta l'\langle n'\{m'_{\sigma(1)}/x_1,\dots,m'_{\sigma(k)}/x_k\}\rangle}\ **$$

$$\frac{m \rightrightarrows_\delta m' \qquad n \rightrightarrows_\delta n' \qquad l \rightrightarrows_\delta l'}{l\langle\lambda x m\rangle n \rightrightarrows_\delta l'\langle m'[n'/x]\rangle} \qquad \frac{m_1 \rightrightarrows_\delta m'_1}{\mathbf{der}([m_1,\dots,m_k]) \rightrightarrows_\delta m'_1}\ ***$$

* s is any surface resource context

** $\sigma \in P_k$, and if $k = d_x(n')$ (otherwise the reduction gives $\emptyset$).

*** $k = 1$. Otherwise, the reduction gives $\emptyset$.

Figure 2: Parallel reduction for δBang

2.2.4 Parallel reduction and full contexts

Figure 2 defines a parallel resource reduction allowing us to handle full contexts. Intuitively, it needs to reduce at once every term occurring in a bag, in order to simulate internal reductions like $!M \to_! !M'$: in this case, approximants of the antecedent are of the form $[m_1,\dots,m_k]$ with $m_i \lhd_! M$, and the simulation needs to reduce all the m_i's at once, in order to obtain an approximation of the reduct $[m'_1,\dots,m'_k] \lhd_! !M'$. In particular, the reduction needs to be reflexive because *e.g.* $[]\lhd_! !M$ and $[]$ does not reduce to $\emptyset$, we have to consider that $[]\lhd_! !M'$ is obtained from $[]$ by reduction. The resulting relation $\rightrightarrows_\delta$, is a well-known non-deterministic extension of standard reduction which can be used to prove confluence property. In general, for a reduction $\to$, we have $\to \subsetneq \rightrightarrows \subsetneq \to^*$[3]. See *e.g.* Barendregt's proof of confluence for the λ-calculus [8], known as Tait—Martin-Löf technique. [4] We abusively write $l \rightrightarrows_\delta l'$ for contexts as soon as $l = \Box[m_1/x_1]\dots[m_k/x_k], l' = \Box[m'_1/x_1]\dots[m'_k/x_k]$ and $m_i \rightrightarrows_\delta m'_i$ for any $i \leq k$.

Since $\rightrightarrows_\delta$ is size-decreasing, it enjoys weak normalization, but obviously not strong, as it is reflexive. We can now state our simulation results for full contexts:

Lemma 19. *Let $M, N \in$ dBang with $M \to_! N$. For any $m \lhd_! M$, either $m \rightrightarrows_\delta \emptyset$ or there is some $n \lhd_! N$ such that $m \rightrightarrows_\delta n$*

The symmetric counterpart of this result is obtained with a similar reasoning:

Lemma 20. *Let $M, N \in$ dBang with $M \to_! N$. For any $n \lhd_! N$, there is some $m \lhd_! M$ such that $m \rightrightarrows_\delta n$.*

The two previous lemmas give us the desired simulation result for full reduction:

Theorem 21. *Let $M, N \in$ dBang such that $M \to_! N$, then we have $\mathcal{T}(M)\overline{\rightrightarrows_\delta}\mathcal{T}(N)$*

We saw that surface reduction acts similarly in dBang and in δBang, while parallel reduction is necessary to give a multilinear account to internal reductions (the definition of parallel reduction alone does not imply that they occur exclusively inside bags, but this is made mandatory through the use of invariant multi-hole surface contexts). The factorization properties established for dBang (Corollary 6) can easily be translated in the resource setting:

Proposition 22 (Factorization)**.** *Let s^+ denote multi-holes surface contexts and given by the syntax $s^+ := \Box \mid m \mid s^+s^+ \mid s^+[s^+/x] \mid \mathbf{der}(s^+) \mid \lambda x s^+$. For any reduction $m \to_\delta^* p$, there are some bags $\bar{n}_i$ and some multi-hole context s such that $p = s\langle\bar{n}_1,\dots,\bar{n}_k\rangle$ and $m \to_{\delta s}^* s\langle\bar{n}'_1,\dots,\bar{n}'_k\rangle \rightrightarrows_\delta^* s\langle\bar{n}_1,\dots,\bar{n}_k\rangle$*

Following the reduction occurring on resource terms, we have defined previously Taylor normal form. We develop in this section some lemmas on those objects. They will be useful in order to prove the Commutation Theorem between approximation trees and Taylor expansions.

Lemma 23. *Given $M \to_!^* N$ then $TNF(M) = TNF(N)$.*

Lemma 24. *Given $m \in TNF(M)$ then there exists M' such that $M \to_!^* M'$ and $m \lhd_! M'$.*

[3] Usually the parallel reduction enjoys the diamond property. It is not the case here because of the non-determinism induced by the permutations; in order to recover it, it is necessary to lift the reduction to set of terms. We mentioned it with non parallel reduction, whith $\Rightarrow_\delta$ being confluent, but not $\to_{\delta s}$, for the same reason.

[4] See Vaux-Auclair [20] for a detailed study of this reduction in λ-calculus and its interaction with quantitative Taylor expansion.

2.3 Approximation trees

A classical way to approximate λ-terms is through evaluation trees, which can be seen as infinitary normal forms. The evaluation tree of a λ-term corresponds to the limit of a sequence of increasingly precise approximants of this λ-term. Those approximants are terms in normal forms, but using potentially a specific subterm $\bot$, representing an undefined computation. When reducing the λ-term some subterms become stable under further reductions, these subterms are precisely those captured by approximants.

Different comparable evaluation trees coexist. The original and best-known ones are Böhm trees. They have been introduced in the CbN paradigm where they intersect with numerous notions. Indeed, they give a precise classification of λ-terms in regard of their operationnal behaviour.

In CbN, using all partial normal forms (possibly containing $\bot$) as approximants, corresponds to Lévy-Longo trees [19]. Böhm trees use a restriction on this set of approximants, so that unsolvable terms have an empty Böhm tree. This restriction is based on restricting approximants to (partial) head normal forms, since being solvable is equivalent to having a head normal form. Lévy-Longo trees instead rely on weak head normal forms. In fact, it only leads to the suppression of "stupid" approximants such as $\lambda x\bot$. Böhm trees and Lévy-Longo trees differ essentially in their treatment of abstractions over undefined computations and therefore share similar properties. Among others, they both satisfy variants of the Commutation Theorem, which states that Taylor expansion of the Böhm tree of a term is equal to its Taylor normal form [16].

More recently, Böhm trees have been introduced in the CbV setting (in a variant with permutation rules) [17]. They are defined without restriction on approximants. Indeed, in this case, there is no known syntactic characterization of solvability in terms of a specific normal form. In that, they may be considered closer to the Lévy-Longo trees. However, the relation with scrutability — the notion of meaning — in CbV is respected: unscrutable λ-terms have an empty Böhm tree. An approximant such as $\lambda x\bot$ is meaningful in this setting, since it is a value.

In this work we define approximation trees for the distant version of CbV, CbN and Bang. As in the CbV case, we do not restrict our approximants for dCbN and dCbV. In order to be consistent between the different calculus, we adopt the same approach for dCbN. The resulting trees are called approximation trees here. They correspond to Lévy-Longo trees in the dCbN case, while in the dCbV case they are closer to Böhm trees in spirit, although the situation is more subtle. Since those two family of trees are very strongly related, we choose not to focus too much on whether our trees should be considered as Böhm or Lévy–Longo trees.

In this section we will introduce those trees for dBang and their relation to the Taylor expansion via the Commutation Theorem (Theorem 46).

Definition 25 ($\mathtt{dBang}_\bot$)**.** *Let* $\mathtt{dBang}_\bot$ *be the set of* dBang*-terms extended with the symbol* $\bot$*. In the following we use subscripts as* $\bot_i$ *in order to distinguish occurrences of* $\bot$ *as a subterm. Similarly, we extend the different types of contexts; and we extend reductions in the obvious way.*

Definition 26. *The set of approximants is a strict subset of* $\mathtt{dBang}_\bot$ *generated by the grammar:*

$$\begin{array}{rcl} A_! & := & B \mid \lambda x A \mid A_![A_!/x] \\ A_\lambda & := & B \mid\, !A \mid A_\lambda[A_!/x] \end{array}$$

$$\begin{array}{rcl} A & := & \bot \mid B \mid \lambda x A \\ B & := & x \mid A_\lambda A \mid \mathbf{d} \end{array}$$

Lemma 27. *Approximants are the normal forms of* $\mathtt{dBang}_\bot$*.*

Substituting $\bot$ by any term preserves normal forms, as a consequence of the syntactic structure of approximants. First, approximants contain no redexes. Secondly, there are no approximants containing subterms of shape $\mathbf{der}(\bot), \bot A, A[\bot/x]$ that could hide a potential redex. Formally:

Lemma 28. *Let* A *be an approximant,* M *any term of* $\mathtt{dBang}_\bot$*, and* $\bot_i$ *some occurrence of* $\bot$ *in* A*. If* $A[M/\bot_i] \to_! N$*, then* $N = A[M'/\bot_i]$ *with* $M \to_! M'$*.* i.e. *we cannot create a redex when replacing a* $\bot$ *by a term* M *in an approximant: the only redexes obtained this way are those already present in* M*.*

Definition 29 (Order)**.** *Let* $\sqsubseteq \subseteq \mathtt{dBang}_\bot \times \mathtt{dBang}_\bot$ *be the least contextual closed preorder on* $\mathtt{dBang}_\bot$ *generated by setting:* $\forall M \in \mathtt{dBang}_\bot$ *and for all full context* F*,* $F[\bot] \subseteq F[M]$*.*

In other words, $M \sqsubseteq N$ if and only if $M = N\overrightarrow{\{P/\bot\}}$. In the evaluation trees literature, we sometimes find approximation orders where only terms of shape $!P$ are replaced by a $\bot$ (see *e.g.* Dufour and Mazza [13]). For technical reasons (Theorem 59 in particular) due to CbN and cbv embeddings into dBang, we need, for example, $\lambda x\bot \sqsubseteq \lambda x M$ to be a valid approximation.

The following lemma states that when a reduction occurs in a term, it cannot be *seen* by its approximations, that only represent a part of the skeleton of their normal form. In other words, when $A \sqsubseteq M$, then A represents a subtree of M which cannot be modified by some reduction.

Lemma 30. *$A \sqsubseteq M$ and $M \to_!^* N$ then $A \sqsubseteq N$.*

Definition 31 (Set of approximants)**.** *Given $M \in \mathtt{dBang}$, the set of approximants of M is defined as follows: $\mathcal{A}(M) = \{A \mid M \to_!^* N, A \sqsubseteq N\}$.*

Note that $\mathcal{A}(M)$ is never empty, since it contains at least $\bot$.

Example 32. *Consider the terms given in Example 3.*

- *The only reducts of Ω being Ω itself, and since the syntax A of approximants contains no redex, we easily conclude that $\mathcal{A}(\Omega) = \{\bot\}$.*
- *The reducts of Y_x^n are of shape $x!x!x \dots !Y_x^n$, we conclude that $\mathcal{A}(Y_x^n)$ contains precisely the terms $\bot, x\bot, x!\bot, x!x\bot, x!x!\bot$*
- *The reducts of Y_x^v follow the same observation, minus the exponentials, and $\mathcal{A}(Y_x^v)$ contains precisely $\bot, x\bot, xx\bot, xxx\bot, \dots$*

Lemma 33. *$M \to_!^* N$ then $\mathcal{A}(M) = \mathcal{A}(N)$.*

Remark 34. *If $A = F[\vec{\bot}] \in \mathcal{A}(M)$ (where F is a multi-hole context), then $M \to^* F[\vec{N}]$ for some $\vec{N}$. In particular, if A contains no $\bot$ subterms, then M has a normal form, which is exactly A.*

Definition 35 (Ideal)**.** *Let $X \subseteq \mathtt{dBang}_\bot$. We set X is an* ideal *when X is downwards closed for $\sqsubseteq$, and directed: for all $M, N \in X$ there exists some upper bound (with respect to $\sqsubseteq$) to $\{M, N\}$.*

Lemma 36. *Let $M \in \mathtt{dBang}$. $\mathcal{A}(M)$ is an ideal.*

In order to define approximation trees, we follow the long-established tradition for term rewriting systems, including CbN and CbV [2, 10, 9, 1, 3, 17], based on the ideal completion. This method constructs the set of ideals of approximants (ordered by a direct partial order). Evaluation trees, such as Böhm trees are then identified with such ideals. The finite and infinite ideals represent respectively the finite and infinite trees. For a λ-term M, its evaluation tree is the ideal generated by its set of approximants; equivalently, it can be seen as the supremum of these approximants in the associated directed-complete domain. This domain admits a concrete presentation as a coinductive grammar extending that of approximants, where constructors may be unfolded infinitely often. Given $M \in \mathtt{dBang}$, $\mathcal{A}(M)$ has a supremum, noted $\cup\mathcal{A}(M)$, which is a potentially infinite tree.

Definition 37 (Approximation Tree)**.** *The approximation tree of a term M in $\mathtt{dBang}$, is given by $\cup\mathcal{A}(M)$, and denoted $\mathtt{AT}_!(M)$.*

Approximation trees satisfy the following immediate properties (we already used these facts on approximants for previous results, they lift immediately to approximation trees).

Proposition 38 (Some properties of approximation trees)**.**

- *If M is in normal form, then $\mathtt{AT}_!(M) = M$.*
- *$\mathtt{AT}_!(M) = \bot$ if and only if $\mathcal{A}(M) = \{\bot\}$[5], if and only if any reduct of M is a redex.*
- *If $M \to^* N$, $\mathtt{AT}_!(M) = \mathtt{AT}_!(N)$, for example $\mathtt{AT}_!(\mathbf{der}(!M)) = \mathtt{AT}_!(M)$.*
- *$\mathtt{AT}_!(\lambda x M) = \lambda x \mathtt{AT}_!(M)$ and $\mathtt{AT}_!(!M) = !(\mathtt{AT}_!(M))$.*
- *If $\mathtt{AT}_!(M) \neq L\langle \lambda x - \rangle$, then $\mathtt{AT}_!(MN) = \mathtt{AT}_!(M)\mathtt{AT}_!(N)$.*
- *If $\mathtt{AT}_!(M) \neq L\langle !- \rangle$, then $\mathtt{AT}_!(\mathbf{der}(M)) = \mathbf{der}(\mathtt{AT}_!(M))$ and $\mathtt{AT}_!(N[M/x]) = \mathtt{AT}_!(N)[\mathtt{AT}_!(M)/x]$*

In particular, we can infer from the facts above that if $\mathtt{AT}_!(M)$ is an application, then it is equal to $(x)\mathtt{AT}_!(N)$ for some N (hence $M \to^* L\langle x \rangle N$).

Example 39. *From Example 32, we can infer that $\mathtt{AT}_!(\Omega) = \bot$, $\mathtt{AT}_!(Y_x^n)$ is the infinite application $x!x!x!x \dots$, and $\mathtt{AT}_!(Y_x^v)$ is also an infinite application $xxxxx \dots$. This is the intended behaviour of approximation trees, as in this category of terms they represent, at the limit, the amount of result produced by a computation, even if the term itself has no normal form.*

[5] This distinguishes our approximation trees from those of Mazza and Dufour [13], in which $\mathtt{AT}_!(M) = \bot$ as soon as M has no surface normal form. Indeed, for technical reasons (relative to CbV embeddings), if $\mathtt{AT}_!(M) = \bot$, we need to have $\mathtt{AT}_!(\lambda x M) = \lambda x \bot$, and not $\bot$, because $\lambda x \bot$ embeds to $!(\lambda x \bot)$, while $\bot$ embeds to $\bot$, which loses the exponential and breaks commutation properties between approximation trees and CbV embedding (Theorem 59). This distinction vanishes at the semantical level: as soon as we consider Taylor expansion of approximation trees, the trees having no surface normal form are given an empty expansion (Definition 40).

2.4 The commutation between approximation tree and Taylor expansion

Combining the previous results, we state the Commutation Theorem. We start by defining the Taylor expansion of approximation trees.

Definition 40. *We extend the Taylor expansion to* $\mathtt{dBang}_\perp$*-terms by setting* $\mathcal{T}(\perp) = \emptyset$ *(recall that* $f[\emptyset] = \emptyset$ *for any full context* f*). In other words, there is no* $\delta\mathtt{Bang}$ *term* m *such that* $m \triangleleft_! \perp$*.*

One immediate consequence of this definition is that, for M in $\mathtt{dBang}_\perp$, $\mathcal{T}(M) \neq \emptyset$ if M contains no $\perp$ as a surface subterm. In other words, M expands to a non-empty set if and only if the $\perp$ are under exponentials ! (in that case, these exponentials can be approximated by empty bags). Moreover, $M \sqsubseteq N$ implies $\mathcal{T}(M) \subseteq \mathcal{T}(N)$.

Definition 41 (Taylor expansion of approximation trees)**.** *Given* $M \in \mathtt{dBang}$*,* $\mathcal{T}(\mathtt{AT}_!(M)) = \cup_{a \in \mathcal{A}(M)} \mathcal{T}(a)$*.*

Example 42. *Following the terms of previous examples, we easily check that* $\mathcal{T}(\mathtt{AT}_!(\Omega)) = \emptyset$ *(because* $\mathcal{A}(\Omega) = \{\perp\}$*). Recall that* $\mathcal{A}(Y_x^n) = \{\perp, x\perp, x!\perp, \dots\}$*. The first of these approximants having a non-empty expansion is* $x!\perp$*, approximated by* $x[]$*,* $\mathcal{T}(\mathtt{AT}_!(Y_x^n)) = \{x[], x[x[], \dots, x[\dots]], \dots\}$*. And* $\mathcal{T}(\mathtt{AT}_!(Y_x^v)) = \emptyset$*. Indeed, all approximants of* Y_x^v *are of shape* $xxx\dots x\perp$*, and have a surface* $\perp$ *that expands to* $\emptyset$*. With examples 32 and 12, we observe for these terms the identity of* $TNF(M)$ *and* $\mathcal{T}(\mathtt{AT}_!(M))$ *which is proved globally in this section.*

Remark that, since $\mathcal{A}(M)$ is an ideal, $\mathcal{T}(\mathtt{AT}_!(M))$ is a directed union. The purpose of Taylor expansion is to approach a term by finitary resource terms only, so we do not consider any infinite supremum of this union, and keep a set of terms, this approximation being inductive; while Böhm trees are coinductive and consist of infinite objects.

The two following lemmas formalize a compatibility between approximation order and inclusion of Taylor expansion.

Lemma 43. *Let* $A \sqsubseteq M$*, then* $\mathcal{T}(A) \subseteq \mathcal{T}(M)$*.*

Lemma 44. *Let* $A \in \mathcal{A}(M)$*, then* $\mathcal{T}(A) \subseteq TNF(M)$*.*

Analogously, normalized resource approximants match the tree approximants.

Lemma 45. *Let* $m \triangleleft_! M$ *in normal form, then there exists an approximant* A *such that* $A \sqsubseteq M$ *and* $m \triangleleft_! A$ *(remember that* $m \triangleleft_! M$ *and* $m \in \mathcal{T}(M)$ *are the same thing).*

From the previous lemmas we deduce the Commutation Theorem:

Theorem 46. *Let* $M \in \mathtt{dBang}$*.* $\mathcal{T}(\mathtt{AT}_!(M)) = TNF(M)$*.*

3 Translations

This section is about the approximation theory of dCbN and dCbV, in particular about how the embeddings into dBang can profit from the result of previous Section. We briefly recall the relevant definitions and results of those embeddings, that have been already well studied [11, 18, 4, 6, 5, 7] dCbV and dCbN share the same syntax: $M, N ::= x \mid \lambda x M \mid MN \mid M[N/x]$. Furthermore, the values V are defined as either a variable x or a λ-abstraction $\lambda x M$. List contexts are defined as in dBang; reduction rules are:

- In dCbN: $L\langle\lambda x M\rangle\ N \to_{\mathrm{n}} L\langle M[N/x]\rangle$ and $M[N/x] \to_{\mathrm{n}} M\{N/x\}$
- In dCbV:$L\langle\lambda x M\rangle\ N \to_{\mathrm{v}} L\langle M[N/x]\rangle$ and $M[L\langle V\rangle/x] \to_{\mathrm{v}} L\langle M\{N/x\}\rangle$

We are now ready to define the translations of dCbN (noted $()^n$) and dCbV (noted $()^v$) into dBang. Several translations of dCbV into dBang exist (or the original Bang-calculus without explicit substitutions). The first one [15], inspired by Girard *second translation*, does not preserve normal forms (xy translates to $\mathbf{der}(!x)!y$). Another translation was then proposed [11], which fixes this problem by simplifying the created redexes by the translations on the fly. This is the translation that we use here. It is worth noticing that this translation does not satisfy reverse simulation from dBang to dCbV [5, Figure 1] This issue has been addressed by a third translation [5]. However, reverse simulation is not necessary in our case, and we will keep with the translation proposed in [11], although we are convinced that the same developments could be carried out with the translation from [5]. This choice is motivated by the fact that the translation we consider fits better with the Linear Logic discipline from which stems Taylor expansion, and also because when considering the study of meaningfulness, we can rely on results that have been proved for this translation. Moreover, this translation enjoys a weaker property than strict reverse simulation, that we call embedding (see Lemma 58) and which is sufficient for our study (approximation trees and Taylor expansion concern mostly iterated reductions $\to^*$, and one-step, thus reverse simulation is then not mandatory for our results).

$$
\begin{aligned}
x^n &= x \\
(\lambda x M)^n &= \lambda x M^n \\
(M\ N)^n &= M^n !N^n \\
(M[N/x])^n &= M^n[!N^n/x]
\end{aligned}
\qquad
\begin{aligned}
x^v &= !x \\
(\lambda x M)^v &= !(\lambda x M^v) \\
(M\ N)^v &= \begin{cases} L\langle P\rangle N^v & \text{if } M^v = L\langle !P\rangle \\ \mathbf{der}(M^v)\ N^v & \text{otherwise} \end{cases} \\
(M[N/x])^v &= M^v[N^v/x]
\end{aligned}
$$

We abusively extend the translations to list contexts: let $\circ \in \{n, v\}$; then given a list context $L = \Box[M_1/x_1]\dots[M_k/x_k]$, we write $L^\circ = \Box[M_1^\circ/x_1]\dots[M_k^\circ/x_k]$.

We are now ready to define meaningfulness in both systems:

Definition 47 (dCbV and dCbN meaningfulness [18])**.** *We say that a term $M \in$ dCbV is dCbV-meaningful (resp. $M \in$ dCbN and is dCbN-meaningful) if $T\langle M\rangle \to_v^* V$ for some value V (resp. $T\langle M\rangle \to_n^* \lambda x x$) where T is a testing context, defined by: $T := \Box \mid T\ N \mid (\lambda x T)\ N$.* [6]*.*

Example 48. *Consider the fixpoint $Y_x = (\lambda y x(yy))(\lambda y x(yy))$, its translations are:*

- $(Y_x)^n = (\lambda y x !(y!y))!(\lambda y x !(y!y))$.
- $(Y_x)^v = (\lambda y x (y!y))!(\lambda y x (y!y))$.

Which are the terms Y_x^n and Y_x^v of Example 3.

It has been shown that dCbN dCbV can be simulated through their encoding in dBang, and that the meaningfulness of dCbV and dCbN coincides with the one of dBang (Definition 4):

Theorem 49.

1. *If $M \to_n N$ (resp. $M \to_v N$) then $M^n \mapsto_! N^n$ (resp. $M^v \mapsto_! N^v$) [11, Lemma 4.6].*
2. *M is dCbN-meaningful iff M^n is meaningful [18, Theorem 25].*
3. *M is dCbV-meaningful iff M^v is meaningful [18, Theorem 30].*

We study approximation trees and Taylor expansion for dCbV and dCbN via their translations, in view of meaningfulness.

3.1 Taylor expansion and approximation trees for dCbN and dCbV

3.1.1 Taylor expansion for dCbN and dCbV

Definition 50 (Resource approximations and Taylor expansion of dCbN)**.**

We define an approximation $\lhd_n$ relation between δBang and dCbN. Notice that despite the approximants being defined in δBang, there is no dereliction needed in the case of dCbN. The relation is given by $x \lhd_n x$; if $m \lhd_n M$ then $\lambda x m \lhd_n \lambda x M$; and if $m \lhd_n M$ and $n_i \lhd_n N$ for any $i \leq k$ then we have $m[n_1, \dots, n_k] \lhd_n MN$ and $m[[n_1, \dots, n_k]/x] \lhd_n M[N/x]$.

Taylor expansion is again defined as sets of approximations: $\mathcal{T}^n(M) = \{m \in \delta\text{Bang} \mid m \lhd_n M\}$.

We do not need to define a specific resource calculus for dCbN nor dCbV, since δBang semantics precisely subsumes both approximation theories.

Remark 51. *The syntax of approximants of dCbN can be given by the following syntax: $m, n := x \mid m\ [n_1, \dots, n_k] \mid \lambda x m \mid n$*

Definition 52 (Resource approximation and Taylor expansion of dCbV)**.** *We define the relation $m \lhd_v M$ for $m \in \delta$Bang and $M \in$ dCbV.*

- $[x, \dots, x]_k \lhd_v x$ *for any* $k \in \mathbb{N}$.
- $[\lambda x m_1, \dots, \lambda x m_k] \lhd_v \lambda x M$ *if* $m_i \lhd_v M$ *for any* $i \leq k$.
- $\mathbf{der}(m) n \lhd_v MN$ *if* $m \lhd_v M, n \lhd_v N$ *and* $M \notin V$
- $mn \lhd_v VN$ *if* $[m] \lhd_v V$ *and* $n \lhd_v N$
- $m[n/x] \lhd_v M[N/x]$ *if* $m \lhd_v M$ *and* $n \lhd_v N$.

[6] Note that as for dBang, if such a reduction exists, then a weak reduction (analoguous to surface reduction) is sufficient, *i.e.* it is not necessary to reduce under the λs in dCbV or in the arguments of applications (or explicit substitutions) in dCbN.

Taylor expansion is defined as $\mathcal{T}^v(M) = \{m \in \delta\texttt{Bang} \mid m \lhd_v M\}$. *Notice that, so as Taylor expansion commutes with this embedding, Taylor approximation also suppresses derelictions redexes, so as the expansion also preserves normal forms.*

Example 53. *Approximants can fit both for* dCbN *and* dCbV *terms, such as* $x[x] \lhd_v xx$ *and* $x[x] \lhd_n xx$. *But this is not the case in general,* e.g. $[x] \lhd_v x$ *but there is no* dCbN *term* M *such that* $[x] \lhd_v M$.

Both in dCbV and dCbN we also define $\mathrm{TNF}(M)$ as the set containing the normal forms of resource approximants of M. We get the following:

Lemma 54. *Let* $M \in$ dCbN *(resp.* dCbV*) then* $\mathcal{T}^n(M) = \mathcal{T}(M^n)$ *(resp.* M^v*)*

In the dCbV case, notice that the translation of application, as well as its Taylor expansion, is described by case on its first component such that the translation (resp. the expansion) of a term in normal form does not lead to any reducible pattern. From that we obtain the following properties:

Corollary 55. *Let* $M \in$ dCbN *(resp.* dCbV*) then* $TNF(M) = \emptyset \leftrightarrow TNF(M^n) = \emptyset$ *(resp.* M^v*).*

3.1.2 Approximation trees for dCbN and dCbV

Definition 56. *The syntax of* dCbN *approximants is given by:*

$$A_n ::= \bot \mid N_\lambda \mid \lambda x A_n \qquad\qquad N_\lambda := x \mid N_\lambda A_n$$

Notice that this syntax contains inductive weak head normal forms, as it is standard for Lévy-Longo trees [19]. Subterms such as $\lambda\vec{x}\bot$, which might correspond to non trivial approximants of non-solvable terms (*e.g.* $\lambda x\Omega$) are not included in Böhm trees, which correspond to inductive head normal forms. We could endow the approximants with equations identifying $\lambda\vec{x}\bot$ to $\bot$, but this is not necessary[7] since this trivialization is achieved by Taylor expansion.

Definition 57. *The syntax of* dCbV *approximants is given by:*

$$\begin{array}{lcl} A_v & ::= & \bot \mid A_\lambda \mid \lambda x A_v \mid A_v[A_{x\lambda}/x] \\ A_\lambda & ::= & x \mid A_\lambda A_v \mid A_\lambda[A_{x\lambda}/x] \end{array} \qquad\qquad A_{x\lambda} ::= A_\lambda A_v \mid A_{x\lambda}[A_{x\lambda}/x]$$

In dCbV, as well as in the CbV case (and its variant with permutation rules), there is no equivalent to (weak) head normal forms. We define our approximants as the normal forms potentially containing $\bot$. This is similar to the ones used for Böhm trees in [17] (for a variant of CbV with permutation rules).

We then define approximation trees in the usual way, setting $\mathtt{AT}_\circ(M) = \cup\{A_\circ \mid M \to_\circ^* N, A_\circ \sqsubseteq N\}$, for $\circ \in \{n, v\}$, (and where $M \sqsubseteq N$ means again that M is obtained by replacing subterms of N by $\bot$). We leave it to the reader to verify that the proof of Lemma 36 can be adapted to dCbV and dCbN, incorporating the distant setting into these standard results.

In the following, we extend translations $()^n$ and $()^v$ to approximants by setting $\bot^n = \bot^v = \bot$, and study the commutation between approximation and the embedding (Theorem 59), that will allow us to transport our Commutation Theorem (Theorem 46) into dCbV and dCbN. Notice that M is a dCbN (resp. dCbV) approximant if and only if M^n (resp. M^v) is a dBang approximant.

Now we can prove the weaker form of reverse simulation mentioned in the preamble of this section. Notice that this result corresponds to the embedding notion of Dufour and Mazza [13].

Lemma 58 (Embedding)**.**

1. *Let* $M \in$ dCbN*. If* $M^n \to_! N$*, then there is some* $P \in$ dCbN *such that* $M \to_n^* P$ *and* $N \to_!^* P^n$.
2. *Let* $M \in$ dCbV*. If* $M^v \to_! N$*, then there is some* $P \in$ dCbV *such that* $M \to_v^* P$ *and* $N \to_!^* P^v$.

Theorem 59.

1. *Let* $M \in$ dCbN*.* $(\mathtt{AT_n}(M))^n = \mathtt{AT}_!(M^n)$.
2. *Let* $M \in$ dCbV*. Then* $(\mathtt{AT_v}(M))^v = \mathtt{AT}_!(M^v)$.

Thanks to the compatibility of approximation trees and Taylor expansion with the translations of dCbN and dCbV into dBang we can apply our commutation result for dBang to both calculi. Although these results have been well established (to our knowledge, only for non-distant CbN and CbV), this application illustrates that the subsuming power of dBang has been brought in the approximation theory, which was our purpose.

Theorem 60.

1. *Let* $M \in$ dCbV*.* $\mathcal{T}^v(\mathtt{AT}_v(M)) = \mathbf{nf}(\mathcal{T}^v(M))$.
2. *Let* $M \in$ dCbN*.* $\mathcal{T}^n(\mathtt{AT}_n(M)) = \mathbf{nf}(\mathcal{T}^n(M))$.

[7] In fact, as for dBang, having $\mathtt{AT}(\lambda x M) = \lambda x \mathtt{AT}(M)$ holding globally is convenient for technical reasons.

4 Meaningfulness and Taylor expansion

In CbN and CbV, solvability and scrutability are characterized by non-empty Taylor normal forms [16, 12]. We aim to establish an analogous link in dBang, for which a characterization of meaningfulness has been provided [18], but without an approximation theory related to this notion. The first part of this result is achieved by Theorem 61. The second part presents signifiant challenges, which this section aims to address.

Theorem 61. *Let $M \in \mathtt{dBang}$. If M is meaningful, then $TNF(M) \neq \emptyset$.*

This result is encouraging for our study of Taylor expansion in dBang framework, as it applies to dCbN and dCbV through our previous results (Corollary 55): Theorem 61 applies to both settings.

The converse fails in general: some non-empty Taylor normal forms correspond to meaningless terms. As mentioned in [18], some elementary terms, such as xx, are meaningless, whereas xy is not. Their intersection type system can distinguish between these terms, but it is unlikely that such a distinction can be captured at the syntactic level using Taylor expansion. Another approach would be to restrict ourselves to a (clash-free) fragment of dBang excluding patterns that do not make sense from a dCbV nor a dCbN discipline (such as xx), but again we can exhibit terms such as $(x!x)(x!x)$ (Recall that $(xM)^v = !x(M^v)$ and $(Mx)^n = M^n!x.$) that are meaningless but cannot reasonably be assigned an empty Taylor normal form.

We prove in the remaining of this section that the result holds independently for the two sublanguages of dBang consisting of terms translated from $()^v$ and $()^n$. Our proof employs techniques adapted from CbN [16] and CbV [12], providing an initial characterization of the relationship between meaningfulness and Taylor expression in a distant setting. Although it is frustrating that we cannot prove the equivalence once for dBang and to apply it directly to its fragments; this limitation also raises an open question which we find to be of interest: is there a significant, bigger fragment of dBang for which the equivalence can be proven generically? Would this fragment cover terms not coming from a dCbV or dCbN translations? This is, for now, an open question.

We consider two strict subsets of dBang: $\mathtt{dBang}_V$ and $\mathtt{dBang}_N$ corresponding to terms obtained by translating from dCbN and dCbV, respectively. These fragments also have the advantage of excluding *clashes* - problematic dBang terms such as $\mathbf{der}(\lambda x M)$ - which are often omitted from the analysis [11, 15] (see Remark 13).

Definition 62.

$\mathtt{dBang}_N$: $M_n := x \mid \lambda x M_n \mid M_n!M_n \mid M_n[!M_n/x]$
$\mathtt{dBang}_V$: $M_v := !x \mid !(\lambda x M_v) \mid L_v\langle \lambda x M_v\rangle M_v \mid L_v\langle x\rangle M_v \mid \mathbf{der}(M_v)M_v \mid M_v[M_v/x]$
$L_v := \Box \mid L_v[M_v/x]$

A simple inspection of the definitions yields the following property:

Lemma 63. *For any $M \in \mathtt{dCbV}$, $M^v \in \mathtt{dBang}_V$, and for any $M \in \mathtt{dCbN}$, $M^n \in \mathtt{dBang}_N$.*

Note that the converse holds for $\mathtt{dBang}_N$, but not for $\mathtt{dBang}_V$. For example $\mathbf{der}(!x)M \in \mathtt{dBang}_V$, but no term of dCbV translates to this term due of the side condition of $(-)^v$ which ensures the preservation of normal forms. However, we cannot exclude these patterns from $\mathtt{dBang}_V$ as they can be obtained from some reduction as shown in Section 3.

We aim to ensure that our fragments are closed under reduction. Otherwise, a term in $\mathtt{dBang}_V$, for example, could reduce in a term in dBang for which meaningfulness cannot be guaranteed (such as xx or clashes like $\mathbf{der}(\lambda x M)$). The following lemma can be proven by a standard induction.

Lemma 64. *$\mathtt{dBang}_V$ and $\mathtt{dBang}_N$ are closed under $\to_!$.*

4.1 Meaningfulness and Taylor Expansion for dCbN

The case of dCbN is relatively easy to handle, as we can adapt to the distant case the following properties; which correspond to well-known features of λ calculus: resource normal form correspond to head normal forms and terms with head normal forms are meaningful.

Lemma 65. *The normal forms of $\mathtt{dBang}_N$ are of shape $\lambda x_1 \dots x_k(x)!N_1 \dots !N_l$, where $k, l \in \mathbb{N}$.*

Naturally, full normal form requires the N_i to be in normal form too, but as we shall see, this is not relevant for studying meaningfulness, as these terms will be erased by an appropriate testing context. Previous observations can be brought at a resource level.

Lemma 66. *Let $m \lhd_n M$ with $M \in \mathtt{dBang}_N$. If $\mathbf{nf}(m) \neq \emptyset$, it is of shape $\lambda x_1 \dots x_k(x)\bar{n}_1 \dots \bar{n}_l$.*

We are now able to state the theorem establishing the classical link between Taylor expansion and meaningfulness in the case of $\mathtt{dBang}_N$.

Theorem 67. *Let $M \in \mathtt{dBang}_N$. If $TNF(M) \neq \emptyset$, then M is meaningful.*

Corollary 68. *For any M in dCbN, $TNF(M) \neq \emptyset$ if and only if M is meaningful.*

4.2 Meaningfulness and Taylor expansion for dCbV

The dCbV case requires a finer analysis of normal forms. We first characterize normal forms of $\mathtt{dBang}_V$ and their counterparts in resource calculus. The following definition and lemma are derived by a standard induction over the syntax of $\mathtt{dBang}_V$.

Definition 69. *The normal forms of* $\mathtt{dBang}_V$ *are described by the following syntax:*

$B := B_! \mid !\lambda x B \mid !x \mid L\langle B\rangle \qquad L := \Box \mid L[B_!/x]$

$B_! := L\langle x\rangle B \mid \mathbf{der}(B_!)B \mid L\langle B_!\rangle$

Lemma 70. *Let* $m \lhd_! M \in \mathtt{dBang}_V$*. If* $\mathbf{nf}(m) \neq \emptyset$ *then it is of the following form:*

$b := b_! \mid [\lambda x b, \dots, \lambda x b] \mid [x, \dots, x] \mid l\langle b\rangle \qquad l := \Box \mid l[b_!/x]$

$b_! := l\langle x\rangle b \mid \mathbf{der}(b_!)b \mid l\langle b_!\rangle$

We consider a family of terms which are suitable for providing an appropriate testing context for any term of $\mathtt{dBang}_V$ with non-empty Taylor normal form, in which it eventually reduces to a value. We define $\circ_0 = \lambda x_0 x_0$ and $\circ_{k+1} = \lambda x_{k+1} !\circ_k$.

We establish the testing context with the following lemma. The proof (see Appendix C) follows Carraro and Guerrieri's technique of mutual induction [12, Lemmas 26 and 27].

Lemma 71. *Let* $\{x_1, \dots, x_n\}$ *be a set of variables and* $M \in B$ *(Definition 69) with* $\mathbf{fv}(M) \subseteq \{x_1, \dots, x_n\}$*. There exists* $c \in \mathbb{N}$ *such that for any* $k_1, \dots, k_n \geq c$ *we have* $M\{\circ_{k_1}/x_1 \cdots \circ_{k_n}/x_n\} \to_!^* !P$ *for some* P*.*

We can now state the central theorem of this section.

Theorem 72. *Let* $M \in \mathtt{dBang}_V$*. If* $TNF(M) \neq \emptyset$*, then* M *is meaningful.*

Corollary 73. *Let* $M \in \mathtt{dCbV}$ *then* M *is meaningful if and only if* $TNF(M) \neq \emptyset$*.*

A Proofs of Section 2

Lemma 74. *If* $M \mapsto_! N$*, then for any* $m \lhd_! M$*, either* $m \to_\delta \emptyset$*, or there is some* $n \lhd_! N$ *such that* $m \to_\delta n$*.*

Proof of Lemma 74. By induction on the definition $\mapsto_!$:

- If $m \lhd_! \mathbf{der}(L\langle !N\rangle)$, then there exist $k \in \mathbb{N}, n_1, \dots, n_k \lhd_! N$, and $l \lhd_! L$ such that $m = \mathbf{der}(l\langle [n_1, \dots, n_k]\rangle)$, and then $m \mapsto_! \emptyset$ if $k \neq 1$, otherwise $m \mapsto_! l\langle n_1\rangle \lhd_! L\langle N\rangle$.
- If $m \lhd_! (L\langle \lambda x N\rangle P)$, then $m = l\langle \lambda x n\rangle p$ for some $l \lhd_! L, n \lhd_! N$ and $p \lhd_! P$. Then, $m \to_\delta l\langle n[p/x]\rangle \lhd_! L\langle N[P/x]\rangle$.
- If $m \lhd_! N[L\langle !P\rangle/x]$ then there are $k \in \mathbb{N}, p_1, \dots, p_k \lhd_! P, n \lhd_! N, l \lhd_! N$ such that $m = n[l\langle [p_1, \dots, p_k]/x]$. Then $m \to_\delta \emptyset$ if $d_x(m) \neq k$, and otherwise for any $\sigma \in P_k$, $m \to_\delta l\langle n\{p_{\sigma(1)}/x_1, \dots, p_{\sigma(k)}/x_k\}\rangle$. □

This simulation property can be extended to surface contexts:

Lemma 75. *If* $M \to_{!s} N$*, then for any* $m \lhd_! M$*, either* $m \to_{\delta s} \emptyset$ *or there is some* $n \lhd_! N$ *such that* $m \to_{\delta s} n$*.*

Proof of Lemma 75. Let $M = S\langle M'\rangle$ and $N = S\langle N'\rangle$ with $M \mapsto_! N$; and then $m = s\langle m'\rangle$ for $s \lhd_! S$. By induction on surface contexts:

- $S = \Box$. Then $M \mapsto_! N$, we apply Lemma 74.
- $S = \lambda x S'$. Then $M = \lambda x S'\langle M'\rangle$ and $N = \lambda x S'\langle N'\rangle$. By induction hypothesis, either $m' \to_{\delta s} \emptyset$, and then $\lambda x s'\langle m'\rangle \to_{\delta s} \emptyset$, either we have some $n' \lhd_! N'$ such that $m' \to_{\delta s} n'$, and then $\lambda x s\langle m'\rangle \to_\delta \lambda x s\langle n'\rangle$ by definition of resource surface reduction.

All remainder cases are similar, since for any resource surface context s and term $n \lhd_! N$, there exist S such that $s\langle n\rangle \lhd_! S\langle N\rangle$ which enables the induction hypothesis. Again, this fails for full contexts. □

We can obtain a symmetric property with the same arguments. We could qualify, following Dufour and Mazza [13], the previous lemma as *push forward*, and the next lemmas as *pull back*.

Lemma 76. *If* $M \to_{!s} N$*, then for any* $n \lhd_! N$*, there is some* $m \lhd_! M$ *such that* $m \to_{\delta s} n$*.*

Proof of Lemma 76. Let $M = S\langle M'\rangle, N = S\langle N'\rangle$ with $M' \mapsto_! N'$ then we can reason by induction on S.

- $S = \Box$. Then $M \mapsto_! N$ and by a straightforward case analysis on $\mapsto_!$.

- $S = \lambda x S'$. Then $M = \lambda x S'\langle M'\rangle, N = \lambda x S'\langle N'\rangle$ and $m = \lambda x s'\langle m'\rangle$ with $s' \lhd_! S'$. By induction hypothesis there exists $m' \lhd_! M'$ such that $m' \to_\delta n'$ then let $m = \lambda x s'\langle m'\rangle$ we have that $m \to_{\delta s} n$ by closure.

All the remainder cases are similar. □

Proof of Lemma 19. We proceed by induction on the exponential depth where the reduction occurs. By lemma 75, and since $\to_{\delta s} \subseteq \rightrightarrows_\delta$, the only case we need to show is the full context closure, where $M = F\langle M'\rangle = !F'\langle M'\rangle$ and $N = F\langle N'\rangle$.

Then, $m = [p_1, \dots, p_k]$ for some $k \in \mathbb{N}$, where $p_i \lhd_! F'\langle M'\rangle$.

By induction hypothesis, either some p_i reduces to $\emptyset$, and then also $m \rightrightarrows_\delta \emptyset$, either for any $i \leq k$, there is some $p'_i \lhd_! F'\langle N'\rangle$ such that $p_i \rightrightarrows_\delta p'_i$. Then, $m \rightrightarrows_\delta [p'_1, \dots, p'_k] \lhd_! !F'\langle N'\rangle = N$. □

Proof of Lemma 23. By Theorem 21, we immediately have $\mathrm{TNF}(M) \subseteq \mathrm{TNF}(N)$. Then, let $n \in \mathrm{TNF}(N)$. By iteration of Lemma 20, there is some $m \in \mathcal{T}(M)$ such that $m \rightrightarrows_\delta^* n$. Then n must be in $\mathrm{TNF}(M)$. □

Proof of Lemma 24. Consider $m_0 \lhd_! M$ such that $m_0 \to_\delta^* m$. We reason by induction over the exponential depth under which the reduction occurs. If this depth is 0, then $m_0 \to_{\delta s}^* m$ and we apply (iteratively) Lemma 15 to conclude.

Otherwise, we use the factorization property : by Proposition 22, we have some n such that $m_0 \to_{\delta s}^* n = s\langle \bar{n}_1, \dots, \bar{n}_k\rangle \rightrightarrows_\delta^* s\langle \bar{n}'_1, \dots, \bar{n}'_k\rangle = m$. By Lemma 15, we have some $N = S\langle !N_1, \dots, !N_k\rangle$ such that $M \to_{!s}^* N$ and $n \lhd_! N$ (thus $n_i \lhd_! N_i$). Then, for any $i \leq k$, $\bar{n}_i = [n_{i,1}, \dots, n_{i,l_i}]$ and $\bar{n}'_i = [n'_{i,1}, \dots, n'_{i,l_i}]$ (in normal form, since these are subterms of m) with $n_{i,j} \lhd_! N_i$ and $n_{i,j} \to_\delta^* n'_{i,j}$. This reduction occurs under an exponential (*i.e.* inside a multiset), we then can apply our induction hypothesis to assert that there is some N'_i such that $N_i \to_! N'_i$ and $n'_{i,j} \lhd_! N'_i$. We can conclude, by setting $M' = S\langle !N'_1, \dots, !N'_k\rangle$. □

A.1 Proof of Section 2.3

Proof of Lemma 30. We show the lemma holds for one-step reductions, assuming $M \to_! N$. The closure is easily obtained by induction on the number of steps. By definition, $M = A\{P_1/\bot_1, \dots, P_k/\bot_k\}$, and by Lemma 28, there must be some $j \in \{1, \dots, k\}$ such that $N = A(\{P_i/\bot_i\})_{i\neq j}\{P'_j/\bot_j\}$ with $P_j \to_! P'_j$. Consequently, we have $A \sqsubseteq N$. □

Proof of Lemma 33. From Definition 31 we deduce $\mathcal{A}(N) \subseteq \mathcal{A}(M)$, and from Lemma 30 the other inclusion. □

Proof of Lemma 36. The downwards closure is by definition of $\mathcal{A}(M)$. For directedness, let us assume $A_1, A_2 \in \mathcal{A}(M)$, we show that there exists $A_3 \in \mathcal{A}(M)$ such that $A_1 \sqsubseteq A_3$ and $A_2 \sqsubseteq A_3$, by induction on A_1.

- If $A_1 = x$ then $M \to_!^* x$ and $A_2 = x$ or $\bot$ (Remark 34), then we set $A_3 = x$.
- If $A_1 = \bot$, by definition of $\sqsubseteq$ we have $A_1 \sqsubseteq A_2$, we set $A_3 = A_2$.
- If $A_1 = \lambda x A'_1$, by Lemma 28 we have $M \to_!^* \lambda x N$ and $A'_1 \in \mathcal{A}(N)$. Then, by Lemma 33 we have $A_2 \in \mathcal{A}(\lambda x N)$ so either $A_2 = \bot$ and we set $A_3 = A_2$ or $\lambda x A'_2$ so $A'_2 \in \mathcal{A}(N)$. Then, by induction hypothesis, there is some A'_3 such that $A'_1 \sqsubseteq A'_3$ and $A'_2 \sqsubseteq A'_3$. We then set $A_3 = \lambda x A'_3$.
- If $A_1 = !A'_1$ we reason as in the previous case.
- If $A_1 = \mathbf{der}(A'_1)$. Then $M \to_!^* \mathbf{der}(N)$ and $A' \in \mathcal{A}(N)$. By Lemma 33 we have $A_2 \in \mathcal{A}(\mathbf{der}(N))$ so either $A_2 = \bot$ or $A_2 = \mathbf{der}(A'_2)$ such that $A'_2 \in \mathbf{der}(A_!)$, and therefore we can use our induction hypothesis to obtain an upper bound A'_3 of A'_1 and A'_2 and set $A_3 = \mathbf{der}(A'_3)$, which is indeed an upper bound of $\{A_1, A_2\}$ by contextual closure of the approximation order.
- If $A_1 = A'_1[A''_1/x]$, then $M \to_!^* N'[N''/x] = N$ with $A'_1 \in \mathcal{A}(N')$ and $A''_1 \in \mathcal{A}(N'')$. A''_1 is not of shape $L\langle !-\rangle$, then neither is N''. Again, $A_2 \in \mathcal{A}(N_1)$, then $A_2 = A'_2[A''_2/x]$ with $A'_2 \sqsubseteq N'$ and $A''_2 \sqsubseteq N''$. By induction hypothesis, we can find A'_3 an upper bound of $\{A'_1, A'_2\}$ and A''_3 an upper bound of $\{A''_1, A''_2\}$. We conclude by setting $A_3 = A'_3[A''_3/x]$.
- If $A_1 = A'_1 A''_1$, we reason similarly, but using the fact that A'_1 cannot be of shape $L\langle \lambda x -\rangle$. □

A.2 Proof of Section 2.4

Proof of Lemma 43. By induction on A. If $A = \bot$ then $\mathcal{T}(A) = \emptyset \subseteq \mathcal{T}(M)$. If $A = x$ then $M = x$ and $\mathcal{T}(A) = \mathcal{T}(M) = \{x\}$. If $A = \lambda x A'$, then $M = \lambda x M'$ with $A' \sqsubseteq M'$. By induction hypothesis, $\mathcal{T}(A') \subseteq \mathcal{T}(M')$. Then, $\mathcal{T}(A) = \{\lambda x a' \mid a' \in \mathcal{T}(A')\} \subseteq \mathcal{T}(M) = \{\lambda x m' \mid m' \in \mathcal{T}(M')\}$. The other cases are treated similarly by routine induction. □

Proof of Lemma 44. We have some M' such that $M \to_!^* M'$ and $A \sqsubseteq M'$. We have that $\mathbf{nf}(\mathcal{T}(M)) = \mathbf{nf}(\mathcal{T}(M'))$ by Lemma 23. By Lemma 43 we have $\mathcal{T}(A) \subseteq \mathcal{T}(M')$. We conclude by observing that terms in $\mathcal{T}(A)$ are in normal form, and that normal terms in $\mathcal{T}(M')$ must also belong to $\mathbf{nf}(\mathcal{T}(M'))$ as they are not affected by any reduction. □

Proof of Lemma 45. By induction on m.

- If $m = x$, then $M = x$ and we set $A = x$.
- If $m = \lambda x n$, then $M = \lambda x N$ with $n \lhd_! N$. Since n must be in normal form, we can apply the induction hypothesis to obtain some $A' \sqsubseteq N$ such that $n \lhd_! A'$. We then set $A = \lambda x A'$.
- If $m = m_1 m_2$, then $M = M_1 M_2$ with $m_i \lhd_! M_i$. Again, by induction hypothesis, we have $A_1 \sqsubseteq M_1$ and $A_2 \sqsubseteq M_2$ with $m_i \lhd_! A_i$. It remains to show that $A_1 A_2$ belongs to the set of approximants described in Definition 26. Notice that m_1 cannot be of shape $l\langle \lambda x — \rangle$, then since $m_1 \lhd_! A_1$, A_1 cannot be a bottom or an abstraction. A simple examination of the syntax of approximants is enough to conclude that $A_1 A_2$ indeed belongs to it.
- If $m = \mathbf{der}(n)$, then we again obtain $n \lhd_! N$ and some $A' \sqsubseteq N$ with $n \lhd_! A$. Since n cannot be of shape $l\langle [-] \rangle$. (since m is normal), we can again check Definition 26 to conclude that $A = \mathbf{der}(A)'$ is an approximant, and that $m \lhd_! A$.
- If $m = n[p/x]$, we reason as for the application case, but using the case that p cannot be of shape $l\langle [-] \rangle$.
- If $m = [n_1, \ldots, n_k]$, then $M = !N$ with $n_i \lhd_! N$ for all $i \in \{1, \ldots, k\}$. Then by induction hypothesis, there is $A_i \sqsubseteq N$ with $n_i \lhd_! A_i$. We then take $A = !A_i$, which works for any i. □

Proof of Theorem 46. We proceed by double inclusion.

- Take $m \in \mathcal{T}(\mathtt{AT}_! M)$, then there exists some $A_0 \in \mathcal{A}(M)$ such that $m \in \mathcal{T}(A_0)$, by Definition 41. There is M_0 such that $M \to_!^* M_0$ and $A_0 \sqsubseteq M_0$. We can therefore apply Lemma 44 to conclude that $m \in \mathrm{TNF}(M_0)$, which is equal to $\mathrm{TNF}(M)$ by Lemma 23.
- Assume $m \in \mathrm{TNF}(M)$. By Lemma 24 there exists M_0 such that $M \to_!^* M_0$ and $m \in \mathcal{T}(M_0)$. By Lemma 45, there is $A \sqsubseteq M_0$ such that $m \in \mathcal{T}(A)$. By definition, $A \in \mathcal{A}(M)$, so we conclude that $m \in \mathcal{T}(\mathtt{AT}_! M)$. □

B Proofs of Section 3

Proof of Lemma 54. For the `dCbN` case: Considering a resource term $m \in \delta\mathtt{Bang}$, we can show that $m \lhd_n M$ (see Definition 50) if and only if $m \lhd_! M^n$ (see Figure 1), by an immediate induction on M.

- x is the only approximation of $x = x^n$.
- $m \lhd_n \lambda x N \in \mathtt{dCbN}$ if and only if $m = \lambda x n$ with $n \lhd_n N$, if and only if (induction hypothesis) $n \lhd_! N^n$, and then if and only if $m \lhd_! \lambda x N^n = (\lambda x N)^n$
- $m \lhd_n NP \in \mathtt{dCbN}$ if and only if $m = n[p_1, \ldots, p_k]$ with $n \lhd_n N, p_i \lhd_! P$ for any $i \leq k$ if and only if (induction hypothesis) $n \lhd_! N^n$ and $p_i \lhd_! P^n$, and then if and only if $m \lhd_! (NP)^n$.
- Case $M = N[P/x]$ is similar to the previous one. □

For the `dCbV` case: By induction on M:

- For any $k \in \mathbb{N}$, $[x]_k \lhd_v x$ and $[x]_k \lhd_! x^v = !x$.
- For any $k \in \mathbb{N}$, $[\lambda x m_1, \ldots, \lambda x m_k] \lhd_v \lambda x M$ iff $m_i \lhd_v M$ for all i iff (induction hypothesis) $m_i \lhd_! M^v$ iff $[\lambda x m_1, \ldots, \lambda x m_k] \lhd_! (\lambda x M)^v = !(\lambda x M^v)$.
- $m \lhd_v NP$.

– Either N is an application. Then $m \lhd_v NP$ iff $m = \mathbf{der}(n)p$ with $n \lhd_v N$ and $p \lhd_v P$ iff (induction hypothesis) $n \lhd_! N^v$ and $p \lhd_! P^v$ iff $\mathbf{der}(n)p \lhd_! (NP)^v$.

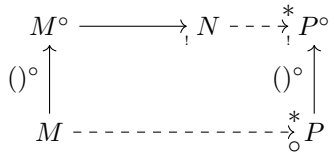
– Either $N = !N'$, and then $m \lhd_v NP$ iff $m = n'p$ with $n' \lhd_v N'$ and $p \lhd_v P$ iff $n' \lhd_! N'^v$, and $p \lhd_! P^v$ iff $n'p \lhd_! (NP)^v$.

Lemma 77 (Substitution)**.**

1. *Let* $M, N \in \mathtt{dCbN}$. $M^n\{N^n/x\} = M\{N/x\}^n$.
2. *Let* $M, N \in \mathtt{dCbV}$. *If* $N^v = !P$, *then* $M\{N/x\}^v = M^v\{P/x\}$

Proof. Standard induction on M. This substitution lemma only holds for value in the case of `dCbV`, because substituting a term to (the traduction of) a variable necessarily puts it under an exponential. This is not an issue because in `dCbV`, these substitutions occur only if it is the case. □

Proof of Theorem 58. For $\circ \in \{v, n\}$, the statements of the lemma can be depicted as follows, where the dashed lines and the term P are the ones to be established:



First, notice that the translations $()^v$ and $()^n$ generate only redexes of shape $L\langle\lambda x M\rangle N$ and $M[L\langle !N\rangle/x]$ [8].
(1) By induction on the reduction $M^n \to_! N$.

- $M^n = L^n\langle\lambda x {N_1}^n\rangle !{N_2}^n$ and $N = L^n\langle {N_1}^n[!{N_2}^n/x]\rangle$. We have $N = (L\langle N_1[N_2/x]\rangle)^n$ by definition of $()^n$. Since $M = L\langle\lambda x N_1\rangle N_2$, we have $M \to_! L\langle N_1[N_2/x]\rangle$, and we are done, setting $P = L\langle N_1[N_2/x]\rangle$.
- $M^n = {N_1}^n[L^n\langle !{N_2}^n\rangle/x]$ and $N = L^n\langle {N_1}^n\{{N_2}^n/x\}\rangle$. By Lemma 77, $N = (L\langle N_1\{N_2/x\}\rangle)^n$, and again we are done, setting $P = L\langle N_1\{N_2/x\}\rangle$, since $M = N_1[L\langle N_2\rangle/x] \to_! L\langle N_1\{N_2/x\}\rangle$.
- The reduction is contextual:
  - $M^n = {N_1}^n !{N_2}^n$ and $N = N_1' !N_2^n$ with $N_1 \to_{\mathrm{n}} N_1'$. By induction hypothesis, there is some P_1 such that $N_1 \to_{\mathrm{n}}^* P_1$ and $N_1' \to_!^* {P_1}^n$. Then we set $P = P_1 N_2$, and we have indeed $M = N_1 N_1 \to_{\mathrm{n}}^* P$ and $N = N_1' !N_2^n \to^* {P_1}^n !{N_2}^n = (P_1 N_2)^n = P^n$.
  - $M^n = {N_1}^n !{N_2}^n$ and $N = {N_1}^n !N_2'$ with $N_2 \to_! N_2'$. By induction hypothesis, we have some P_2 such that $N_2 \to_{\mathrm{n}}^* P_2$ and $N_2' \to_!^* {P_2}^n$. We then set $P = N_1 P_2$.
  - $M^n = {N_1}^n[!{N_2}^n/x], N = {N_1}^n[!N_2'/x]$ with $N_2 \to_! N_2'$. We reason as in the previous case.
  - $M^n = \lambda x N_0^n, N = \lambda x N_0'$ with ${N_0}^n \to_! N_0'$. By induction hypothesis, there is P_0 such that $N_0 \to_{\mathrm{n}}^* P_0$ and $N_0' \to_!^* {P_0}^n$. We then set $P = \lambda x P_0$.

(2) By induction on the reduction $M^v \to_! N$. The first two cases are similar to before except the position of the exponential.

- $M^v = L^v\langle\lambda x {N_1}^v\rangle {N_2}^v$ and $N = L^v\langle {N_1}^v[{N_2}^v/x]\rangle$. We have $N = (L\langle N_1[N_2/x]\rangle)^v$ by definition of $()^v$, and $M = L\langle\lambda x N_1\rangle N_2$. We set $P = L\langle N_1[N_2/x]\rangle$, satisfying $M \to_{\mathrm{v}} P$ and $N \to_!^0 P^v$.
- $M^v = {N_1}^v[L^v\langle !{N_2}^v\rangle/x]$ and $N = L^v\langle {N_1}^v\{{N_2}^v/x\}\rangle$, then $M = N_1[L\langle N_2'\rangle/x]$ with N_2' being a variable or an application. By Lemma 77 ${N_1}\{N_2'/x\}^v = {N_1}^v\{{N_2}^v/x\}$, so $N = (L\langle N_1\{N_2/x\}\rangle)^v$. We then set $P = L\langle N_1\{N_2/x\}\rangle$.
- The reduction is contextual. We only detail the case where the reduction occurs in the left member of an application and under a dereliction; the other cases follow from an application of the induction hypothesis as before. The second of these two cases is important, as it is the responsible for the only configuration where P must be distinct from N.
  - $M^v = \mathbf{der}(N_1^v)N_2^v$ (in that case $M = N_1 N_2$ with N_1 not being of shape $L\langle V\rangle$ for any value V) and $N = \mathbf{der}(N_1')N_2{}^v$. By induction hypothesis, there is some P_1 such that $N_1 \to_{\mathrm{v}}^* P_1$ and $N_1' \to_!^* {P_1}^v$. We then have two possibilities:
    * ${P_1}^v \neq L\langle !-\rangle$ (P_1 is not a value). Then, $(P_1 N_2)^v = \mathbf{der}({P_1}^v){N_2}^v$. We then set $P = P_1 N_2$, and we have $M = N_1 N_2 \to_{\mathrm{v}}^* P_1 N_2$ and $N \to_!^* \mathbf{der}({P_1}^v){N_2}^v = P^v$.

[8] Redexes like $\mathbf{der}(L\langle !N\rangle)$ can however appear during reductions, from translations $()^v$, but not in the translation itself.

* ${P_1}^v = L^v\langle !Q^v\rangle$. Then, $(P_1N_2)^v = L^v\langle Q^v\rangle N_2^v$. We have $N \to_!^* \mathbf{der}(L^v\langle !Q^v\rangle)N_2{}^v \to_! L^v\langle Q^v\rangle {N_2}^v$ by a single reduction step[9]. We then set $P = L\langle Q\rangle N_2$. It verifies $N \to_!^* P^v$ as we just saw. We also have $M \to_{\mathtt{v}}^* P$, because $M = N_1N_2$ and $N_1 \to_{\mathtt{v}}^* P_1$. Since ${P_1}^v = L^v\langle !Q^v\rangle$, it follows that Q is a value (either a variable or an abstraction) and that $P_1 = L\langle Q\rangle$, by definition of $()^v$. □

Proof of Theorem 59. For this proof we will benefit from the properties of Böhm trees stated in Proposition 38.

(1) We proceed by coinduction on $\mathtt{AT_n}(M)$.

- If $\mathtt{AT_n}(M) = \bot$, then $\mathcal{A}(M) = \{\bot\}$. We need to show that $\mathcal{A}(M^n) = \{\bot\}$. Consider $A \in \mathcal{A}(M^n)$, we have $M^n \to_!^* N$ with $A \sqsubseteq N$. By Lemma 58, we have some P such that $M \to_!^* P$ and $N \to_!^* P^n$. By Lemma 30, $A \sqsubseteq P^n$. Now, observe that P must be some $\mathtt{dCbN}$ redex, otherwise $\mathcal{A}(M)$ would contain other approximations than $\bot$. Then, by simulation (Theorem 49), P^n also is a redex, and since the syntax of approximants contains no redex, necessarily $A = \bot$. We conclude that $\mathtt{AT_!}(M^n) = \bot$.
- If $\mathtt{AT_n}(M) = x$, then $\mathtt{AT_n}(M)^n = x = \mathtt{AT_!}(x) = \mathtt{AT_!}(x^n)$.
- If $\mathtt{AT_n}(M) = \lambda x\mathtt{AT_n}(N)$, then $(\mathtt{AT_n}(M))^n = (\lambda x\mathtt{AT_n}(N))^n = \lambda x(\mathtt{AT_n}(N))^n$ (by definition of $()^n$). By coinduction hypothesis, $(\mathtt{AT_n}(N))^n = \mathtt{AT_!}(N^n)$. Then, $(\mathtt{AT_n}(M))^n = \lambda x\mathtt{AT_!}(N^n) = \mathtt{AT_!}(\lambda xN^n) = \mathtt{AT_!}((\lambda xN)^n) = \mathtt{AT_!}(M^n)$.
- If $\mathtt{AT_n}(M)$ is an application, then it is equal to some $(x)\mathtt{AT_n}(N)$. Then, we have $(\mathtt{AT_n}(M))^n = (x)!(\mathtt{AT_n}(N))^n = (x)!\mathtt{AT_!}(N^n)$, by coinduction hypothesis, which is equal to $\mathtt{AT_!}((x)!N^n) = \mathtt{AT_!}((x)N^n) = \mathtt{AT_!}(M^n)$.
- $\mathtt{AT_n}(M)$ cannot contain any explicit substitution, as they always correspond to redexes in $\mathtt{dCbN}$.

(2) We proceed by coinduction on $\mathtt{AT_v}(M)$.

- If $\mathtt{AT_v}(M) = \bot$, we reason as above, using this time the second item of Lemma 58.
- If $\mathtt{AT_v}(M) = x$, then $(\mathtt{AT_v}(M))^v = x^v = !x = \mathtt{AT_!}(!x) = \mathtt{AT_!}(x^v)$.
- If $\mathtt{AT_v}(M) = \lambda x\mathtt{AT_v}(N)$, then $(\mathtt{AT_v}(M))^v = !(\lambda x(\mathtt{AT_v}(N))^v)$. By coinduction hypothesis, it is equal to $!(\lambda x\mathtt{AT_!}(N^v)) = \mathtt{AT_!}(!(\lambda xN^v)) = \mathtt{AT_!}((\lambda xN)^v) = \mathtt{AT_!}(M^v)$.
- If $\mathtt{AT_v}(M)$ is an application, then it must be equal to some $x([\mathtt{AT_v}(N_i)/y_i])_{1\le i\le k}\mathtt{AT_v}(N_0^v)$ (because in this case M reduces to some $L\langle x\rangle N_0$).

  Then $(\mathtt{AT_v}(M))^v = x([(\mathtt{AT_v}(N_i))^v/y_i])_{1\le i\le k}(\mathtt{AT_v}(N_0))^v$. By coinduction hypothesis, $(\mathtt{AT_v}(N_j))^v = \mathtt{AT_!}({N_j}^v)$ for $j \in \{0,\dots,k\}$. Then $(\mathtt{AT_v}(M))^v = x([(\mathtt{AT_!}({N_i}^v))/y_i])_{1\le i\le k}\mathtt{AT_!}({N_0}^v)$. Again, for $i \in \{1,\dots,k\}$, $\mathtt{AT_!}({N_i}^v)$ cannot be an exponential, since those explicit substitutions must not be reducible. Then, $(\mathtt{AT_v}(M))^v = \mathtt{AT_!}\left(x\left([{N_i}^v/x]\right)_{i\in\{1,\dots,k\}} N_0^v\right) = \mathtt{AT_!}(M^v)$.
- If $\mathtt{AT_v}(M) = \mathtt{AT_v}(N_1)[\mathtt{AT_v}(N_2)/x]$, then again $\mathtt{AT_v}(N_2)$ cannot be a value, hence $(\mathtt{AT_v}(M))^v = \mathtt{AT_!}({N_1}^v)[\mathtt{AT_!}({N_2}^v)/x]$ (by coinduction hypothesis) $= \mathtt{AT_!}({N_1}^v[{N_2}^v/x]) = \mathtt{AT_!}((N_1[N_2/x])^v)$. □

Proof of Theorem 60. Let $\circ \in \{n, v\}$. We have the following equalities:

$$\begin{aligned}
\mathbf{nf}(\mathcal{T}^\circ(M)) &= \mathbf{nf}(\mathcal{T}(M^\circ)) && \text{By Lemma 54}\\
&= \mathcal{T}(BT(M^\circ)) && \text{By Theorem 46}\\
&= \mathcal{T}((BT_\circ(M))^\circ) && \text{By Theorem 59}\\
&= \mathcal{T}_\circ(BT_\circ(M)) && \text{By Lemma 54}
\end{aligned}$$

□

C Proofs of Section 4

Proof of Theorem 61. We show the contrapositive of the statement: assume that $\mathrm{TNF}(M) = \emptyset$. By definition, for every $m \in \mathcal{T}(M)$, we have $m \to_\delta^* \emptyset$. Consider now any resource testing context t. We can easily establish that $t\langle m\rangle \to_\delta^* \emptyset$; since $\emptyset m = \emptyset$, $(\lambda x\emptyset)m = \emptyset$, and then by induction.

For M to be meaningful, there must exist a testing context T such that $T\langle M\rangle \to_{!s}^k !P$ for some P and $k \in \mathbb{N}$. We have $[] \lhd_! !P$. By iteratively applying Lemma 76, there is some term $s \lhd_! T\langle M\rangle$ such that $s \to_{\delta s}^k []$. By Lemma 9, and because testing contexts are included in surface contexts, we have some $t \lhd_! T, m \lhd_! M$ such that $s = t\langle m\rangle$.

This leads to a contradiction: $t\langle m\rangle \to_{\delta s} []$, yet we have shown that $t\langle m\rangle \to_{\delta s} \emptyset$ for any t. □

[9] These steps are called *administrative* in Arrial, Guerrieri and Kessner's work [6].

Proof of Lemma 65. The following observations suffice:

- terms in $\mathtt{dBang}_N$ containing explicit substitutions are reducible.
- If the leftmost subterm of an application is not a variable, the entire term is reducible. □

Proof of Theorem 67. Consider some $p \in \mathrm{TNF}(M)$, assumed non-empty. Then there is some $m \lhd_! M$ such that $m \to_\delta^* p$.

By Lemma 66, $p = \lambda x_1 \dots x_k(x)\bar{n}_1 \dots \bar{n}_l$ for some $k, l \in \mathbb{N}$. Proposition 22 allows us to focus on surface reduction: there are some $m' = \lambda x_1 \dots x_k(x)\bar{n}'_1, \dots, \bar{n}'_l$ such that $m \to_{\delta s}^* m' \rightrightarrows_\delta^* p$ (the second part of the reduction acting inside the bags).

Then, by iteratively applying Lemma 15, we obtain $M' \in \mathtt{dBang}$ such that $m' \lhd_! M'$ and $M \to_{!s}^* M'$. By the definition of the approximation relation $\lhd_!$, we have $M' = \lambda x_1 \dots x_k(x)!N'_1 \dots !N'_l$ where $\bar{n}'_i \lhd_! !N'_i$.

We now define the appropriate testing context $T = ((\lambda x \Box)!(\lambda y_1 \dots y_l!z_0))!z_1 \dots !z_k$ where the z_i are chosen distinct from the x_j and y_j.

We observe that $T\langle M' \rangle \to_{!s} (\lambda x_1 \dots x_k(\lambda y_1 \dots y_l!z_0)!z_1 \dots !z_k)!N'_1 \dots !N'_k \to_{!s}^2 !z_0$. We conclude as follows: since $T\langle M' \rangle \to_{!s}^2 !z_0$, $M \to_{!s}^* M'$ and T is a surface context, we have $T\langle M \rangle \to_{!s}^* !z_0$ by the contextual closure of $\to_{!s}$. Therefore, M is meaningful. □

Proof of Corollary 68. Recall that, by Corollary 55 and Theorem 49, $\mathrm{TNF}(M) \neq \emptyset$ if and only if $\mathrm{TNF}(M^n) \neq \emptyset$, and M is meaningful if and only if M^n is meaningful.

($\to$) Assume M is meaningful, then M^n is also meaningful, and by Theorem 61, $\mathrm{TNF}(M^n) \neq \emptyset$. It follows that $\mathrm{TNF}(M) \neq \emptyset$.

($\leftarrow$) If $\mathrm{TNF}(M) \neq \emptyset$, then $\mathrm{TNF}(M^n) \neq \emptyset$. By Lemma 63, $M^n \in \mathtt{dBang}_N$, and Theorem 67 implies that M^n must be meaningful. Therefore, M is also meaningful. □

In what follows, the variables in $\circ_i$ are always taken fresh, so as they do not interfere with variables in the terms where the $\circ_i$ are substituted. In particular, we use the fact that for any $i > 1$ and any M, $\circ_k!M \to_!^2 !\circ_{k-1}$.

Proof of Lemma 71. By induction on the syntax B:

- If M is of the form $!x$ or $!\lambda x B$, then we are done since $M\sigma \to_!^0 !P$ for some P and any substitution σ.
- If $M \in B_!$, we apply Lemma 78, which guarantees the existence of a substitution σ such that $M\sigma \to_!^* !\circ_j$ for some j.
- If $M = N[P_1/x_1] \dots [P_k/x_k]$, then $N \in B$ and $P_i \in B_!$ for all i. Let $\{y_1, \dots, y_l\} = \mathbf{fv}(N) \cup \bigcup_{i \leq k} \mathbf{fv}(P_k)$. By induction hypothesis, we have c such that for any $k_1, \dots, k_l \geq c$, $N\{\circ_{k_1}/y_1, \dots, \circ k_l/y_l\} \to_! !P$ for some P. By Lemma 78, for $i \leq k$, there exist c_i and n_i such that for any $k_{i,1}, \dots, k_{i,l} \geq c_i$, and for some $j_i \geq n_i$, $P_i\{\circ_{k_{i,1}}/y_1, \dots, \circ_{k_{i,l}}/y_l\} \to_!^* !\circ_{j_i}$.

  Consider then $m_i = \max\{k_i, k_{i,1}, \dots k_{i,k}\}$ for any $i \leq l$. We then have some P' and some r_i with $N[P_1/x_1] \dots [P_k/x_k]\{\circ_{m_1}/y_1, \dots, \circ_{m_l}/y_l\} \to_!^* !P'[!\circ_{r_1}/y_1] \dots [!\circ_{r_l}/y_l] \to_!^l !P'\{!\circ_{r_1}/y_1\} \dots \{!\circ_{r_l}/y_l\}$, which is again a value as required. □

Lemma 78. *Let $\{x_1, \dots, x_n\}$ be a set of variables and $M \in B_!$ (Definition 69) with $\mathbf{fv}(M) \subseteq \{x_1, \dots, x_n\}$. There exist $k, c \in \mathbb{N}$ such that for any $k_1, \dots, k_n \geq c$, there is some $j \geq k$ with $M\{\circ_{k_1}/x_1 \cdots \circ_{k_n}/x_n\} \to_!^* !\circ_j$.*

Proof. By induction on $B_!$:

- $M = L\langle x \rangle N = (x[N_1/y_1] \dots [N_m/y_m])N$ with $N \in B$, $N_i \in B_!$ for all $x \leq m$, and $\{x_1, \dots, x_n\} = \mathbf{fv}(M)$. By Lemma 71, there exists k such that $N\{\circ_{k_1}/x_1, \dots, \circ_{k_n}/x_n\} \to_!^* !P$ for any $k_i \geq k$ and some P.

  By induction hypothesis, we have for each $i \leq m$, some l_i and c_i such that for any $l_{i,1}, \dots, l_{i,n} \geq l_i$ $N_i\{\circ_{l_{i,1}}/y_1, \dots, \circ_{l_{i,n}}/y_n\} \to_!^* !\circ_{j_i}$ for some $j_i \geq c_i$.

  Let n_x be the index of x in $\{x_1, \dots, x_n\}$ (of course, $x \in \mathbf{fv}(M)$).

  We then set $r_i = \max\{k_i, l_{i,1}, \dots, l_{i,n}\}$ for each $i \neq n_x$; and we consider r_{n_x} an arbitrary integer greater or equal to $\max\{k_{n_x}, l_{n_x,1}, \dots, l_{n_x,n}\}$.

  We find that $M\{\circ_{r_1}/x_1, \dots, \circ_{r_m}/x_m\} \to_!^* \circ_{n_x}[!\circ_{r'_1}/x_1] \dots [!\circ_{r'_m}/x_1]!P'$, with $r'_i \geq r_i$ for all $i \leq m$. The reduction then yields $\circ_{n_x}!P'$, which reduces immediately to $!\circ_{r_{n_x}-1}$. This concludes the case, as the reduction holds for any $r_{n_x} \geq \max\{k_{n_x}, l_{n_x,1}, \dots, l_{n_x,n}\}$.

- $M = \mathbf{der}(N)N'$ with $N \in B_!$, $N' \in B$, and $\{x_1, \dots, x_m\} = \mathbf{fv}(M)$. By induction hypothesis, we have some n, n', c such that for any $n_1, \dots, n_m \geq n$ and , $N\{\circ_{n_1}/x_1, \dots, \circ_{n_m}/x_m\} \to_!^* !\circ_j$ for all $j \geq c$, and for any $n'_1, \dots, n'_m \geq n'$, $N'\{\circ_{n'_1}/x_1, \dots, \circ_{n'_m}/x_m\} \to_!^* !P$ for some P.

  Then, consider $k_i = \max\{n_i, n'_i\}$ for any $i \leq m$, we have that $M\{\circ_{k_1}/x_1, \dots, \circ_{k_m}/x_m\} \to_!^* \mathbf{der}(!\circ_j)!P' \to_! \circ_j!P' \to_!^2 !\circ_{j-1}$ (notice that we need here to take $j > 1$, which is allowed by our hypothesis).

- $M = N[P_1/x_1]\dots[P_k/x_k]$. This case is similar to the third case of Lemma 71: the explicit substitutions are removed after an application of the induction hypothesis. □

Proof of Theorem 72. Consider some $p \in \mathrm{TNF}(M)$, assumed non-empty. There is some $m \lhd_! M$ such that $m \to_\delta^* p$. By Lemma 70, p belongs to the syntax b. Proposition 22 ensures that there is some $p' \in b$ such that $m \to_{\delta s} p'$ (as in Theorem 67, we focus on internal reduction).

By iteratively applying Lemma 76, we obtain P' such that $M \to_!^* P'$ and $p' \lhd_! P'$. By the definition of $\lhd_!$, we also have that $P' \in B$.

Let $\{x_1, \dots, x_k\} = \mathbf{fv}(P')$. Lemma 71 implies that there are some terms $N_1, \dots, N_k$ such that $P'\{N_1/x_1, \dots, N_k/x_k\} \to_!^* !Q$ for some term Q.

We define the testing context $C = (\lambda x_1 \dots \lambda x_k \Box)!N_1 \dots !N_k$, which satisfies $C\langle P' \rangle \to_!^* !Q$. We can conclude that M is meaningful by the contextuality of reduction, since $C\langle M \rangle \to_!^* C\langle P' \rangle \to_!^* !Q$. □

Proof of Corollary 73. Recall that, by Corollary 55 and Theorem 49, $\mathrm{TNF}(M) \neq \emptyset$ if and only if $\mathrm{TNF}(M^v) \neq \emptyset$, and M is meaningful if and only if M^v is meaningful.

($\to$) Assume M is meaningful, then M^v is meaningful, and by Theorem 61, $\mathrm{TNF}(M^v) \neq \emptyset$. It follows that $\mathrm{TNF}(M) \neq \emptyset$.

($\leftarrow$) If $\mathrm{TNF}(M) \neq \emptyset$, then $\mathrm{TNF}(M^v) \neq \emptyset$. By Lemma 63, $M^v \in \mathtt{dBang}_V$, and Theorem 72 implies that M^v is meaningful. Therefore, M is also meaningful. □

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